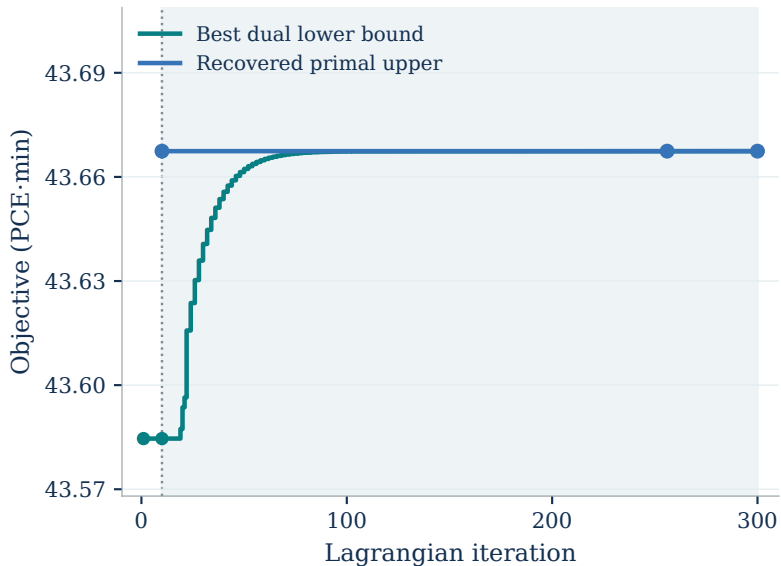
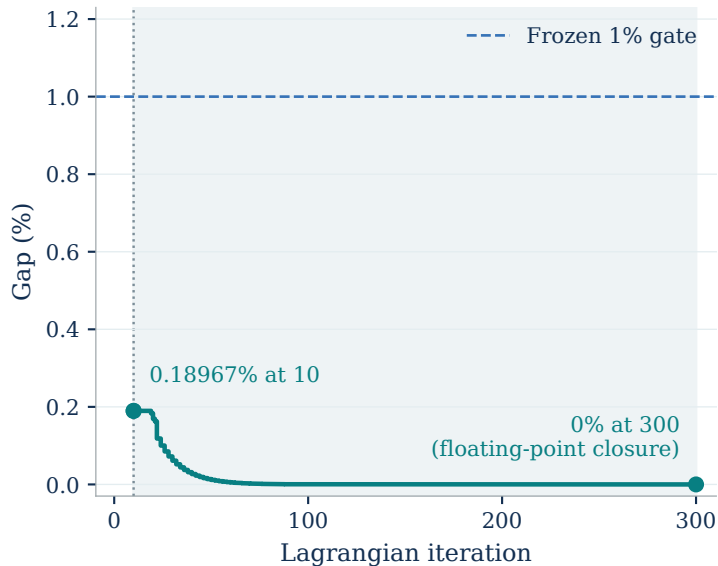


Lagrangian bounds and certified gap

a Best saved bounds**b Certified relative gap**

Dotted line: first formal certificate at iteration 10. Shading: subsequent diagnostic rounds.